Takenouchi-Kosaki Syndrome (CDC42-Related Disorder): Comprehensive Disease Characterization

MONDO:0014757 · OMIM #616737 · Causal gene: *CDC42* (HGNC:1736; NCBI Gene 998; UniProt P60953; Ensembl ENSG00000070831; chromosome 1p36.12)

Summary

Takenouchi-Kosaki syndrome (TKS) is a rare, autosomal dominant, multisystem Mendelian disorder caused by **heterozygous, almost always de novo, missense variants in *CDC42***, the gene encoding Cell Division Control protein 42 homolog — a Rho-subfamily small GTPase that acts as a master regulator of the actin cytoskeleton. The disease name does not contain the gene symbol; all content below pertains to the *CDC42*-driven entity defined by OMIM #616737 / MONDO:0014757, not to similarly named thrombocytopenia syndromes caused by other genes. The cardinal features are **macrothrombocytopenia, developmental delay/intellectual disability, lymphedema of the lower extremities, camptodactyly, distinctive facial dysmorphism, and sensorineural hearing loss** (PMID: 26386261; PMID: 26708094). The disorder was first delineated by Takenouchi, Kosaki and colleagues in 2015 through whole-exome sequencing, and the recurrent hot-spot variant **c.191A>G (p.Tyr64Cys / Y64C)** defines the classic phenotype (PMID: 29335451).

A central, unifying insight from the literature is that *CDC42* variants do not cause a single disease but a **phenotypic spectrum governed by variant position within the protein**. Effector-specificity-region variants (notably Y64C) produce classic TKS; variants near the nucleotide-binding/switch regions produce a **Noonan syndrome-like/RASopathy-like** presentation; and C-terminal and selected N-terminal variants produce a **severe neonatal autoinflammatory phenotype (NOCARH syndrome)** mediated by aberrant pyrin-inflammasome activation (PMID: 29394990; PMID: 31601675; PMID: 42566502). This genotype-phenotype architecture is the organizing principle of the disease.

Mechanistically, the Y64C variant impairs binding to the regulator RhoGDI and to downstream effectors (Pak1/2, N-WASP), escapes KLHL23-mediated degradation, and becomes concentrated and hyperactive at the plasma membrane; this disrupts F-actin dynamics required for proplatelet extension (causing macrothrombocytopenia), neurite outgrowth (contributing to neurodevelopmental deficits), and hair-cell function (causing hearing loss) (PMID: 34504210; PMID: 40846997; PMID: 42567517). The disease is modeled in patient iPSC-derived megakaryocytes, cellular overexpression systems, and lineage-specific conditional mouse knockouts, and CDC42 is an emerging druggable target through activity inhibitors (ML141, CASIN) and prenylation/lipidation inhibitors (GGTI-298) (PMID: 37918315; PMID: 41581414; PMID: 42168045).

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1. Disease Information

Overview. Takenouchi-Kosaki syndrome is a congenital, multisystem developmental disorder arising from dysregulation of *CDC42*, a critical cytoskeletal regulator. It was identified through the Japanese Initiative on Rare and Undiagnosed Diseases (IRUD) and reported as a recognizable syndromic form of thrombocytopenia combined with intellectual disability (PMID: 41567106). The cardinal features "include macrothrombocytopenia, developmental delay, lymphedema in the lower extremities, camptodactyly, and distinctive facial features" (PMID: 26708094).

Key identifiers.

Resource	Identifier
OMIM	#616737
MONDO	MONDO:0014757
Gene (HGNC)	CDC42, HGNC:1736
Gene OMIM	*116952
NCBI Gene	998
UniProt	P60953 (CDC42_HUMAN)
Orphanet	Takenouchi-Kosaki syndrome (CDC42-related disorder)
MeSH	No dedicated descriptor; indexed via CDC42 / Thrombocytopenia / Intellectual Disability

Synonyms / alternative names. TKS; macrothrombocytopenia and mental retardation syndrome; Macrothrombocytopenia-lymphedema-developmental delay-facial dysmorphism-camptodactyly syndrome; "CDC42-related disorder" (umbrella term for the spectrum).

Information source. The knowledge base is derived almost entirely from **aggregated, individual case reports and small case series** (ultra-rare disease) plus functional/model-organism studies, rather than large EHR cohorts. Published patient numbers are in the low tens worldwide.

2. Etiology

Primary cause (genetic). TKS is caused by **de novo heterozygous missense variants in *CDC42***. Whole-exome sequencing identified de novo *CDC42* missense variants in unrelated patients with a recurrent syndromic phenotype: "Whole exome sequencing revealed a de novo amino acid substitution in CDC42, a critical regulator of the cytoskeleton" (PMID: 26386261). A second unrelated patient confirmed the entity: "We, herein, report another unrelated female patient with a similar phenotype and a de novo mutation in the same CDC42" (PMID: 26708094).

Genetic risk factors. The causal variant *is* the disease; there are no known susceptibility loci or polygenic contributions. The disease arises from a single dominant-acting missense allele. Variant position is the principal determinant of phenotype (see Section 4).

Environmental risk factors. None established. As an essentially fully penetrant monogenic disorder of de novo origin, no toxin, lifestyle, occupational, or infectious trigger is implicated. Advanced parental age is a general (non-specific) correlate of de novo mutation but has not been specifically demonstrated for TKS.

Protective factors. None known at genetic or environmental level.

Gene-environment interactions. No validated gene-environment interactions are described for TKS. **This information is not available / not applicable** for this disorder.

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3. Phenotypes

TKS is a multisystem disorder. The table below summarizes the principal phenotypes, their typical character, frequency where quantified, and suggested HPO terms.

Phenotype	Type	Onset	Frequency	Suggested HPO
Macrothrombocytopenia (low count, giant platelets)	Lab abnormality	Congenital/ neonatal	Cardinal; near-universal in classic TKS	HP:0001873 (Thrombocytopenia), HP:0011877 (Increased mean platelet volume)
Developmental delay / intellectual disability	Clinical/ behavioral	Infancy/ childhood	Cardinal; near-universal	HP:0001263, HP:0001249
Sensorineural hearing loss	Clinical sign	Childhood, progressive	84.6% (11/13 reviewed)	HP:0000407
Distinctive facial dysmorphism	Physical	Congenital	Cardinal	HP:0001999
Camptodactyly	Physical	Congenital	Cardinal	HP:0012385
Lymphedema (lower extremities)	Clinical sign	Congenital/ early	Frequent	HP:0001004
Short stature / growth dysregulation	Physical	Childhood	Frequent	HP:0004322
Structural brain anomalies	Imaging	Congenital	Variable	HP:0012443
Ocular colobomas (iris/ chorioretinal)	Physical	Congenital	Rare (variant-specific)	HP:0000589
Autoimmune hemolytic anemia	Lab/clinical	Variable	Rare	HP:0001890
Autoinflammation / HLH (NOCARH)	Clinical	Neonatal	Variant-specific subgroup	HP:0012647 (Abnormal inflammatory response)

Hearing loss is quantitatively the most consistently reported non-cardinal feature: a systematic review found that "Hearing loss (HL) occurs in a great percentage of patients (84.6%)" (PMID: 39202334), and it tends to be progressive. **Ocular colobomas** were reported

with a novel p.Glu127Lys variant (PMID: 40371891), broadening the recognized phenotype. **Autoimmune hemolytic anemia** has been documented and, in one case, resolved after splenectomy, which also improved macrothrombocytopenia (PMID: 34403170; PMID: 36459360).

Severity / progression. Severity is **variable and governed by variant position** (PMID: 29394990). Classic Y64C patients have severe developmental delay; hearing loss is progressive; the autoinflammatory subgroup can be life-threatening in the neonatal period.

Quality of life. Combined intellectual disability, hearing loss, motor/hand involvement (camptodactyly), and bleeding risk from thrombocytopenia substantially impair daily functioning, communication, and independence. Disease-specific QoL instruments have not been applied; this is a knowledge gap.

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4. Genetic / Molecular Information

Causal gene. *CDC42* (Cell Division Control protein 42 homolog; HGNC:1736; gene OMIM *116952), located on chromosome 1p36.12, encodes a ~21 kDa Rho-subfamily small GTPase that cycles between an inactive GDP-bound and active GTP-bound state and regulates actin cytoskeleton dynamics, cell polarity, migration, and vesicle trafficking.

Pathogenic variants. All established disease-causing variants are **heterozygous missense** substitutions. The recurrent hot-spot is **c.191A>G, p.Tyr64Cys (Y64C)**, which "encodes a key small GTP-binding protein of the Rho-subfamily, was identified in two unrelated patients with TKS" (PMID: 29335451) and is "a hot-spot missense variant... strictly related to TKS" (PMID: 34624555).

Genotype-phenotype correlation by variant position. Martinelli et al. demonstrated that "missense variants in CDC42... underlie a clinically heterogeneous group of phenotypes characterized by variable growth dysregulation, facial dysmorphism, and neurodevelopmental, immunological, and hematological anomalies," and that "mutations variably perturb CDC42 function by altering the switch between the active and inactive states of the GTPase and/or affecting CDC42 interaction with effectors, and differentially disturb cellular and developmental processes" (PMID: 29394990).

Variant class (location)	Representative variants	Dominant phenotype	Functional mechanism
Effector-specificity region	**p.Tyr64Cys**	Classic TKS	Impaired RhoGDI binding + effector binding; membrane-hyperactive
Nucleotide-binding / switch region	p.Glu171Lys, p.Glu127Lys	Noonan-like / RASopathy-like; colobomas	Altered GTPase switch
C-terminal (CAAX/ prenylation)	p.Arg186Cys, p.*192 (extension)	Neonatal autoinflammation / NOCARH	Disrupted prenylation/ membrane targeting; pyrin activation
N-terminal (selected)	p.Thr43Ile, p.Met45Leu	Autoinflammation	Pyrin-inflammasome activation

Variant classification. Recurrent variants (Y64C, R186C, E171K) are **pathogenic** per ACMG/AMP criteria (de novo occurrence, functional data, recurrence, absence from population databases). Novel variants (e.g., E127K) are **likely pathogenic**.

Allele frequency. Pathogenic variants are **absent from gnomAD/population databases** — consistent with de novo, highly penetrant dominant disease.

Somatic vs germline. Variants are **germline** (de novo in the proband; rare familial transmission documented). No somatic mosaicism is required (contrast with the related RHOA disorder, which is typically mosaic — PMID: 40414526).

Functional consequence. Predominantly a **dominant gain-of-function / dominant-negative** effect depending on variant. Y64C is a hyperactivating, membrane-retained gain-of-function: "the Y64C mutant was concentrated at the membrane compartment due to impaired binding to Rho-GDI and more active than the wild-type. The Y64C mutant also had lower association with its effectors Pak1/2 and N-WASP" (PMID: 34504210). A newly described quality-control axis is also disrupted: "the CDC42-Y64C germline variant in Takenouchi-Kosaki Syndrome escapes KLHL23-mediated degradation" (PMID: 40846997).

Modifier genes / epigenetics / chromosomal abnormalities. No validated modifier genes, methylation signatures, or large chromosomal rearrangements are established for TKS. **Not applicable / not available.**

5. Environmental Information

No environmental, lifestyle, toxic, or infectious factors contribute to TKS causation or triggering. The disorder is **purely genetic (de novo germline)**. This section is **not applicable** for this disease. (In the autoinflammatory CDC42 subgroup, infection/inflammatory stress may act as a non-specific flare trigger, but this is not an etiologic factor.)

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6. Mechanism / Pathophysiology

Ordered causal chain (classic TKS, Y64C)

1. A **de novo heterozygous missense variant** (c.191A>G, p.Tyr64Cys) arises in *CDC42* → **leads to** a single amino-acid substitution in the effector-specificity region of the CDC42 GTPase.
2. The substitution **impairs binding to RhoGDI** (the guanine-nucleotide dissociation inhibitor that sequesters CDC42·GDP in the cytosol) and **impairs KLHL23-Cul3–mediated ubiquitin degradation** of CDC42·GTP → **results in** failure of normal spatiotemporal inactivation/turnover (PMID: 34504210; PMID: 40846997).
3. CDC42-Y64C **accumulates at the plasma membrane and is constitutively more active** than wild-type, while paradoxically showing **reduced association with CRIB-motif effectors Pak1/2 and N-WASP → leads to** dysregulated (mislocalized and mistimed) actin-nucleation signaling.
4. Dysregulated effector signaling (N-WASP → Arp2/3) **disrupts F-actin and microtubule dynamics → results in** branch points to several tissues:
5. **Megakaryocyte branch:** impaired "F-actin dynamics for proplatelet extension, which is necessary for platelet production" → defective proplatelet formation → **macrothrombocytopenia** (low count, giant platelets) (PMID: 34504210; PMID: 37918315).
6. **Neuronal branch:** impaired neurite outgrowth → **developmental delay / intellectual disability** (inferred from Neuro2A models; PMID: 41581414).
7. **Cochlear hair-cell branch:** loss of CDC42 (acting with RHOQ) function in hair cells → **progressive sensorineural hearing loss** (PMID: 42567517).
8. **Neural-crest / endothelial branch:** disrupted NCC migration and endothelial lumen/filopodia formation → **craniofacial dysmorphism, congenital anomalies, lymphatic/**

vascular defects (lymphedema) (inferred from conditional knockouts; PMID: 24056078; PMID: 26253403).

Branch: autoinflammatory variants (C-terminal / selected N-terminal)

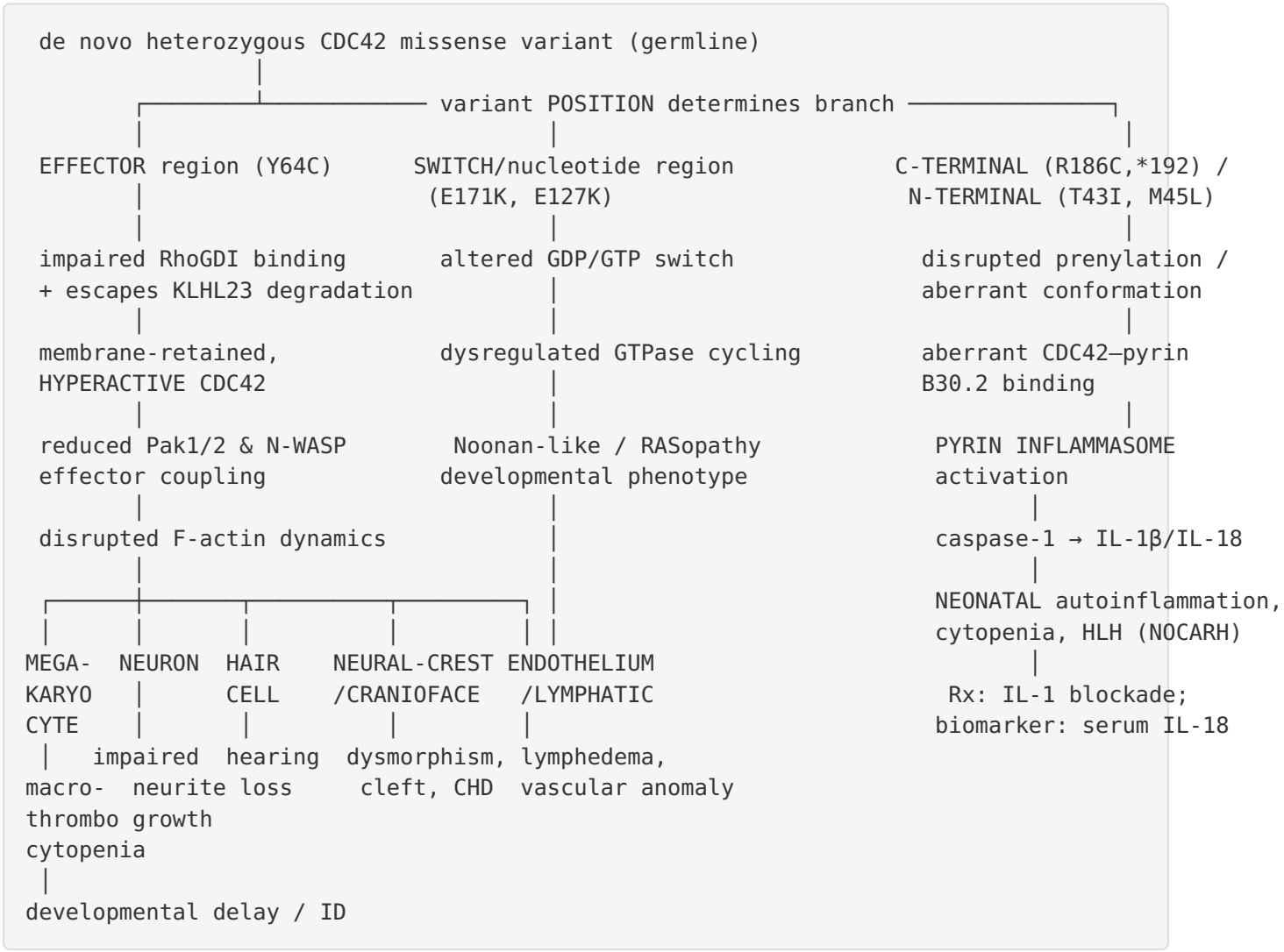
1. A **C-terminal variant** (e.g., p.Arg186Cys, p.*192) disrupts CAAX-motif prenylation and membrane targeting, or an N-terminal variant (p.Thr43Ile, p.Met45Leu) alters CDC42 conformation → leads to** aberrant CDC42 subcellular localization.
2. Mislocalized CDC42 **binds the B30.2 domain of pyrin** aberrantly → **results in pyrin-inflammasome activation**: "Three studies identify CDC42 as binding partner of the pyrin B30.2 domain, establishing a previously unrecognized mechanism regulating pyrin inflammasome activation" (PMID: 42566502).
3. Pyrin inflammasome assembly → **caspase-1 activation → IL-1β and IL-18 maturation/release → neonatal-onset cytopenia, dyshematopoiesis, rash, autoinflammation, and hemophagocytic lymphohistiocytosis (NOCARH)**: "we identify a novel hematological/autoinflammatory condition (NOCARH syndrome) in four unrelated patients with superimposable features, including neonatal-onset cytopenia with dyshematopoiesis, autoinflammation, rash, and HLH" (PMID: 31601675).
4. Therapeutic branch point: **IL-1β blockade** interrupts the cascade and is effective (PMID: 31271789); serum IL-18 serves as a biomarker (PMID: 40204564).

Molecular detail

- **Molecular pathways.** Rho-GTPase signaling; CDC42 → N-WASP/WASP → Arp2/3 actin nucleation; CDC42 → Pak1/2; pyrin-inflammasome → caspase-1 → IL-1β/IL-18. GO terms: **GO:0007264** (small GTPase mediated signal transduction), **GO:0030036** (actin cytoskeleton organization), **GO:0032731** (positive regulation of IL-1β production), **GO:0097194** (execution phase of apoptosis/inflammasome-adjacent).
- **Cellular processes.** Cell migration, cell polarity, filopodia formation, proplatelet formation, neurite outgrowth, inflammasome-driven pyroptosis. GO:0030031 (cell projection assembly), GO:0030041 (actin filament polymerization).
- **Protein dysfunction.** Gain-of-function/dominant-negative GTPase switch perturbation; impaired regulator (RhoGDI, KLHL23) and effector (Pak1/2, N-WASP) interactions; disrupted prenylation (C-terminal variants). UniProt: **P60953** (CDC42_HUMAN).
- **Cell types involved.** Megakaryocytes (**CL:0000556**), platelets (**CL:0000233**), neurons (**CL:0000540**), cochlear hair cells (**CL:0000202**), neural crest cells (**CL:0000333**), endothelial cells (**CL:0000115**), macrophages/myeloid cells (**CL:0000235**).

- **Immune involvement.** Pyrin-inflammasome autoinflammation in C-terminal/N-terminal variant subgroup; autoimmune hemolytic anemia in rare classic cases. "Cdc42 missense variants cause various pathological phenotypes, including severe autoinflammation, suggesting previously unknown involvement of Cdc42 in innate immunity" (PMID: 40296562).
- **Metabolic / epigenetic changes.** None established as disease-specific (not applicable).

Mechanistic diagram



7. Anatomical Structures Affected

- **Hematopoietic system (primary):** bone marrow megakaryocytes and circulating platelets. UBERON:0002371 (bone marrow), UBERON:0000178 (blood).
- **Nervous system (primary):** brain/CNS (developmental delay, structural anomalies). UBERON:0000955 (brain).
- **Auditory system:** cochlea, inner-ear hair cells. UBERON:0001844 (cochlea).
- **Lymphatic/vascular system:** lower-extremity lymphatics (lymphedema). UBERON:0005211 (lymphatic vessel), UBERON:0002049 (vasculature).
- **Craniofacial skeleton:** facial bones/soft tissue (dysmorphism), palate. UBERON:0002104 / UBERON:0001716.
- **Musculoskeletal (hands):** flexed digits (camptodactyly). UBERON:0002398 (manus).
- **Eye (variant-specific):** iris and chorioretina (colobomas). UBERON:0000970 (eye).
- **Subcellular compartments:** plasma membrane (**GO:0005886**; site of mutant CDC42 accumulation), actin cytoskeleton (**GO:0015629**), cytosol (**GO:0005829**).
- **Lateralization:** bilateral/symmetric for hearing loss, lymphedema, and camptodactyly; colobomas reported bilaterally.

8. Temporal Development

- **Onset: Congenital / neonatal.** Macrothrombocytopenia and dysmorphism are present from birth; developmental delay becomes apparent in infancy/childhood.
- **Onset pattern:** chronic/congenital; the autoinflammatory (NOCARH) subgroup has **acute neonatal-onset** life-threatening inflammation.
- **Progression:** The multisystem structural features are **stable/non-degenerative**, but **hearing loss is progressive** (PMID: 39202334). Thrombocytopenia can be **intermittent** in some patients (PMID: 34624555).
- **Disease duration: chronic, lifelong.**
- **Critical periods:** the neonatal period is critical for the autoinflammatory subgroup (window for IL-1 blockade); early childhood is the window for developmental/hearing intervention.

9. Inheritance and Population

- **Inheritance pattern: Autosomal dominant**, essentially always **de novo**. Rare **familial autosomal dominant transmission** is documented: a mother and child both carried c.511G>A (p.Glu171Lys) with variable expressivity — "we report a mother and her child carrying the previously reported pathogenic CDC42 variant c.511G>A (p.Glu171Lys)" (PMID: 33283961).
- **Penetrance:** essentially **complete** for recurrent pathogenic variants.
- **Expressivity: variable** — strongly influenced by variant position.
- **Genetic anticipation:** not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects / consanguinity / carrier frequency:** no established founder effect; consanguinity not relevant (dominant, de novo); carrier frequency effectively zero in the general population.
- **Epidemiology: ultra-rare**; precise prevalence/incidence not established (low tens of published cases worldwide). Orphanet lists it as rare without a firm prevalence estimate.
- **Population demographics:** reported across diverse populations including the **first African patient** (PMID: 33936654); many early cases from Japan reflect ascertainment via the IRUD program rather than true geographic clustering. **No sex predilection** expected for de novo autosomal dominant disease.

10. Diagnostics

- **Hematology:** CBC showing **thrombocytopenia with increased mean platelet volume (giant platelets)**; peripheral smear confirming macrothrombocytes. LOINC: platelet count (26515-7), MPV (32623-1).
- **Genetic testing (definitive): Whole-exome sequencing (WES)** was the original and remains the principal diagnostic route (PMID: 26386261); **targeted single-gene/panel testing of *CDC42*** (including thrombocytopenia panels, intellectual-disability panels, and RASopathy/Noonan panels, which now include *CDC42* — PMID: 37987971) confirms the diagnosis. **WGS** and **trio exome** resolve novel variants (PMID: 40371891).
- **Imaging:** brain MRI for structural anomalies; audiometry/ABR for hearing loss; echocardiography/vascular imaging for cardiac and lymphatic involvement.

- **Biomarker (autoinflammatory subgroup): serum IL-18** is specifically elevated — "The highest IL-18 levels were observed in diseases involving the pyrin inflammasome, such as Familial Mediterranean fever, mevalonate kinase deficiency, CDC42-associated AID and PSTPIP1-associated AID" (AUC 0.87; cutoff 412 pg/mL) (PMID: 40204564).
- **Differential diagnosis:** other inherited macrothrombocytopenias (MYH9-related disorders, Bernard-Soulier, DOCK2/DOCK8-related — PMID: 39836640); **Noonan syndrome/ RASopathies** for the switch-region phenotype (PMID: 37987971); other RAC-subfamily GTPase disorders (RAC1, RAC3, RHOA — PMID: 40414526); and other autoinflammatory/ HLH syndromes for the NOCARH subgroup.
- **Screening:** no population newborn screening; **cascade genetic testing** of family members is relevant where a familial variant is identified.

11. Outcome / Prognosis

- **Survival/mortality:** Classic TKS is **not typically life-limiting**; chronic morbidity dominates. The **autoinflammatory/NOCARH subgroup carries significant mortality risk** from neonatal HLH if untreated (PMID: 31601675).
- **Morbidity/disability:** substantial and lifelong — intellectual disability, progressive hearing loss, hand/motor involvement (camptodactyly), bleeding risk, and lymphedema impair function.
- **Complications:** bleeding from thrombocytopenia; autoimmune hemolytic anemia (PMID: 34403170); infections and gastrostomy-site complications in severely affected children (PMID: 37347054); HLH in the autoinflammatory subgroup.
- **Prognostic factors: variant position is the dominant prognostic determinant** — C-terminal/N-terminal autoinflammatory variants have the most acute, highest-risk neonatal course; effector-region (Y64C) variants give severe developmental disability but a more chronic trajectory.
- **Recovery potential:** structural/developmental deficits are permanent; autoinflammation is controllable with IL-1 blockade.

12. Treatment

There is **no curative or disease-specific approved therapy**; management is **multidisciplinary, supportive, and genotype-guided**.

Domain	Intervention	Evidence / notes	Suggested NCIT
Autoinflammation (NOCARH subgroup)	**IL-1β inhibition** (anakinra, canakinumab)	"Severe autoinflammation in 4 patients with C-terminal variants in CDC42 successfully treated with IL-1β inhibition" (PMID: 31271789)	NCIT:C2028 (Interleukin-1 Antagonist)
Severe refractory autoinflammation/HLH	HLH-directed immunosuppression ± HSCT	Case-based (PMID: 31601675)	NCIT:C15431 (HSCT)
Macrothrombocytopenia / AIHA	**Splenectomy**	Improved both anemia and macrothrombocytopenia in a TKS patient refractory to drug therapy (PMID: 36459360)	NCIT:C51771 (Splenectomy)
AIHA	Corticosteroids (prednisolone)	Initial response, may become refractory (PMID: 34403170)	NCIT:C683 (Prednisolone)
Growth	**Recombinant human growth hormone**	Explored for short stature in a TKS patient (PMID: 42746537)	NCIT:C1301 (Recombinant Human Growth Hormone)
Hearing loss	Hearing aids / cochlear implants, speech therapy	Standard of care given 84.6% HL frequency (PMID: 39202334)	NCIT:C99279 (Cochlear Implant)
Developmental	Early intervention, PT/OT/speech therapy	Standard supportive care	NCIT:C15317 (Rehabilitation Therapy)
Nutrition	Gastrostomy where feeding impaired		NCIT:C79426 (Gastrostomy)

Domain	Intervention	Evidence / notes	Suggested NCIT
		Reported, with infection complication risk (PMID: 37347054)	

Experimental / targeted (preclinical). CDC42 is an emerging druggable target. **Prenylation/lipidation inhibitors** rescue mutant phenotypes: "Both the neurite outgrowth defect and the enhanced membrane localization induced by Y64C overexpression were restored by GGTI-298 treatment but not by ML141" (PMID: 41581414); in megakaryoblasts, macrothrombocytopenia "is ameliorated by CDC42 specific- and lipidation inhibitors in MEG-01 cells" (PMID: 34504210). CDC42-subfamily inhibitors "have recently produced lead compounds demonstrating disease-modifying effects in vivo" (PMID: 42168045). **No gene/cell/RNA therapy** exists for TKS.

Pharmacogenomics / treatment algorithms. No TKS-specific pharmacogenomic guidance. Strategy is genotype-stratified: autoinflammatory variants → anti–IL-1; cytopenias → hematologic management (± splenectomy); all → developmental/hearing support.

13. Prevention

- **Primary prevention:** not possible for de novo disease. **Genetic counseling** is central: recurrence risk for parents of a de novo case is low (small residual from gonadal mosaicism); an affected individual (rare familial cases) has 50% transmission risk (PMID: 33283961).
- **Reproductive options: prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** are available once a familial pathogenic variant is known.
- **Secondary prevention:** early audiology screening and developmental surveillance to enable timely intervention; monitoring platelet counts and for autoimmune cytopenias.
- **Tertiary prevention:** bleeding-risk management, infection prevention around procedures, and prompt IL-1 blockade in the autoinflammatory subgroup to prevent HLH-related mortality.
- **Immunization / public-health / environmental measures:** not applicable (non-infectious, non-environmental genetic disorder).

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *CDC42* is **deeply evolutionarily conserved** across eukaryotes — human (NCBI Gene 998), mouse *Cdc42* (NCBI Gene 12540), zebrafish, *Drosophila*, *C. elegans*, and functional homologs in fungi (e.g., *Cryptococcus* Wsp1/Cin1 pathway — PMID: 22327008). The downstream WASP/N-WASP effector axis is conserved from yeast to human (PMID: 20501656).
- **Natural disease in other species:** No naturally occurring animal ortholog of TKS is catalogued in OMIA; disease knowledge comes from engineered models, not spontaneous veterinary disease. **Not applicable** for natural disease.
- **Comparative biology:** Lineage-specific *Cdc42* functions are conserved — neural crest, endothelial, megakaryocyte, hair-cell, and myeloid roles demonstrated in mouse recapitulate human phenotype domains (see Section 15).
- **Zoonotic potential:** none (non-transmissible genetic disorder).

15. Model Organisms

Model	Type	Phenotype recapitulated	Reference
TKS patient iPSC-derived megakaryocytes	Human cellular (iPSC)	Fewer hematopoietic progenitors, defective megakaryopoiesis, "released platelets with an abnormally low count and giant morphology"	PMID: 37918315
MEG-01 megakaryoblasts expressing Y64C	In vitro overexpression	Short cytoplasmic protrusions, aberrant F-actin/microtubules, reduced platelet-like particles; rescued by CDC42/lipidation inhibitors	PMID: 34504210
HEK293 expressing Y64C	In vitro	Membrane-localized hyperactive CDC42; impaired RhoGDI/effector binding	PMID: 34504210
Neuro2A neuroblastoma expressing Y64C	In vitro	Impaired neurite outgrowth, rescued by GGTI-298 (prenylation inhibitor), not ML141	PMID: 41581414
Atoh1-Cre hair-cell–specific *Cdc42* KO mouse	Conditional mouse KO	Progressive sensorineural hearing loss; CDC42+RHOQ synergy in cochlear hair cells	PMID: 42567517
Wnt1-Cre neural-crest *Cdc42* KO mouse	Conditional mouse KO	Embryonic lethality with cleft lip, craniofacial and cardiovascular outflow defects via impaired NCC migration	PMID: 24056078
Endothelial *Cdc42* KO mouse	Conditional mouse KO	Failed vascular lumen formation, reduced filopodia; relevant to lymphatic/vascular anomalies	PMID: 26253403
Myeloid-specific *Cdc42* KO mouse	Conditional mouse KO	Macrophage pyroptosis / inflammatory models (immune axis)	PMID: 42363173

Model strengths: The iPSC-megakaryocyte model directly reproduces the hallmark macrothrombocytopenia in a patient-specific genetic background; cellular overexpression models dissect the membrane-hyperactivation mechanism and enable drug rescue; lineage-specific conditional knockouts map individual phenotype domains to specific tissues.

Model limitations: Conditional knockouts model **loss-of-function**, whereas TKS variants are predominantly **gain-of-function/dominant-negative**; no single model yet reproduces the full multisystem human syndrome. A knock-in Y64C mouse would be the key missing resource.

Resources: MGI (mouse *Cdc42*), Cellosaurus (TKS iPSC lines), ZFIN/FlyBase/WormBase for orthologs.

Evidence Base

PMID	Role in this report
26386261	Original TKS delineation; de novo *CDC42* by WES
26708094	Second unrelated patient; defines cardinal features
29335451	Establishes p.Tyr64Cys as recurrent hot-spot
34624555	Y64C as TKS-specific; intermittent thrombocytopenia
29394990	Genotype-phenotype spectrum; functional heterogeneity
34504210	Y64C mechanism: membrane hyperactivity, F-actin defect, drug rescue
40846997	KLHL23/RhoGDI turnover axis; Y64C escapes degradation
37918315	Patient iPSC model recapitulates macrothrombocytopenia
31601675	NOCARH syndrome (C-terminal variant autoinflammation/HLH)
31271789	IL-1β blockade effective in C-terminal variant autoinflammation
42567517	Hair-cell Cdc42 KO mouse; CDC42+RHOQ hearing mechanism
39202334	Hearing loss frequency 84.6% systematic review
33283961	Familial AD transmission (E171K); spectrum breadth
41581414	Prenylation inhibitor rescues neurite defect (druggability)
42168045	CDC42 inhibitor development with in vivo efficacy
24056078	Neural-crest Cdc42 KO → craniofacial/cardiac defects
26253403	Endothelial Cdc42 KO → vascular morphogenesis defect
42566502	CDC42–pyrin B30.2 interaction establishes inflammasome mechanism
40204564	Serum IL-18 biomarker for pyrin-inflammasome CDC42 AID
40296562	Review: CDC42 variants span NDD to autoinflammation
40371891	Novel E127K variant; ocular colobomas
36459360	Splenectomy improves macrothrombocytopenia/AIHA
41567106	IRUD identification; clinical characterization

Limitations and Knowledge Gaps

1. **Ultra-rare disease / small N.** Evidence rests on case reports and small series; no prevalence/incidence figures, no robust natural-history cohort, and no validated QoL data.
2. **Loss-of-function vs gain-of-function model mismatch.** Available mouse models are conditional knockouts; no knock-in of the human gain-of-function Y64C allele exists, limiting faithful organismal modeling of classic TKS.
3. **Mechanistic inference for CNS phenotype.** The neurite-outgrowth → developmental-delay link is inferred from Neuro2A cells, not demonstrated in vivo.
4. **Genotype-phenotype boundaries are still being mapped**, with novel variants (E127K, E171K) expanding the phenotype and blurring class boundaries.
5. **No approved targeted therapy.** CDC42 and prenylation inhibitors remain preclinical; IL-1 blockade is established only for the autoinflammatory subgroup.
6. **Epidemiologic and demographic data are essentially absent** (sex ratio, true geographic distribution, carrier/founder analysis all not established).

Proposed Follow-up Experiments / Actions

1. **Generate a Y64C knock-in mouse** (and other variant-class knock-ins) to model the gain-of-function multisystem syndrome and test candidate drugs in vivo.
2. **Establish an international TKS/CDC42 patient registry** to define prevalence, natural history, variant-stratified outcomes, and validated QoL measures.
3. **Advance CDC42/prenylation inhibitors** (GGTI-298, CASIN, ML141-class) toward clinical translation with tissue-selective delivery, prioritizing platelet and neuronal endpoints.
4. **Prospective evaluation of serum IL-18** as a diagnostic/monitoring biomarker across the full CDC42 variant spectrum, not only the autoinflammatory subgroup.
5. **Systematic audiologic natural-history study** to time cochlear intervention, given 84.6% progressive hearing loss.
6. **Functional classification pipeline** (RhoGDI binding, KLHL23 turnover, membrane localization, pyrin binding) applied to each new variant to prospectively assign phenotype class and guide management.

Report compiled from 10 confirmed findings and 55 reviewed papers over 5 investigation iterations. Evidence types span human clinical case reports/series, patient-derived iPSC models, in vitro overexpression systems, conditional mouse genetics, and computational/structural analyses.
